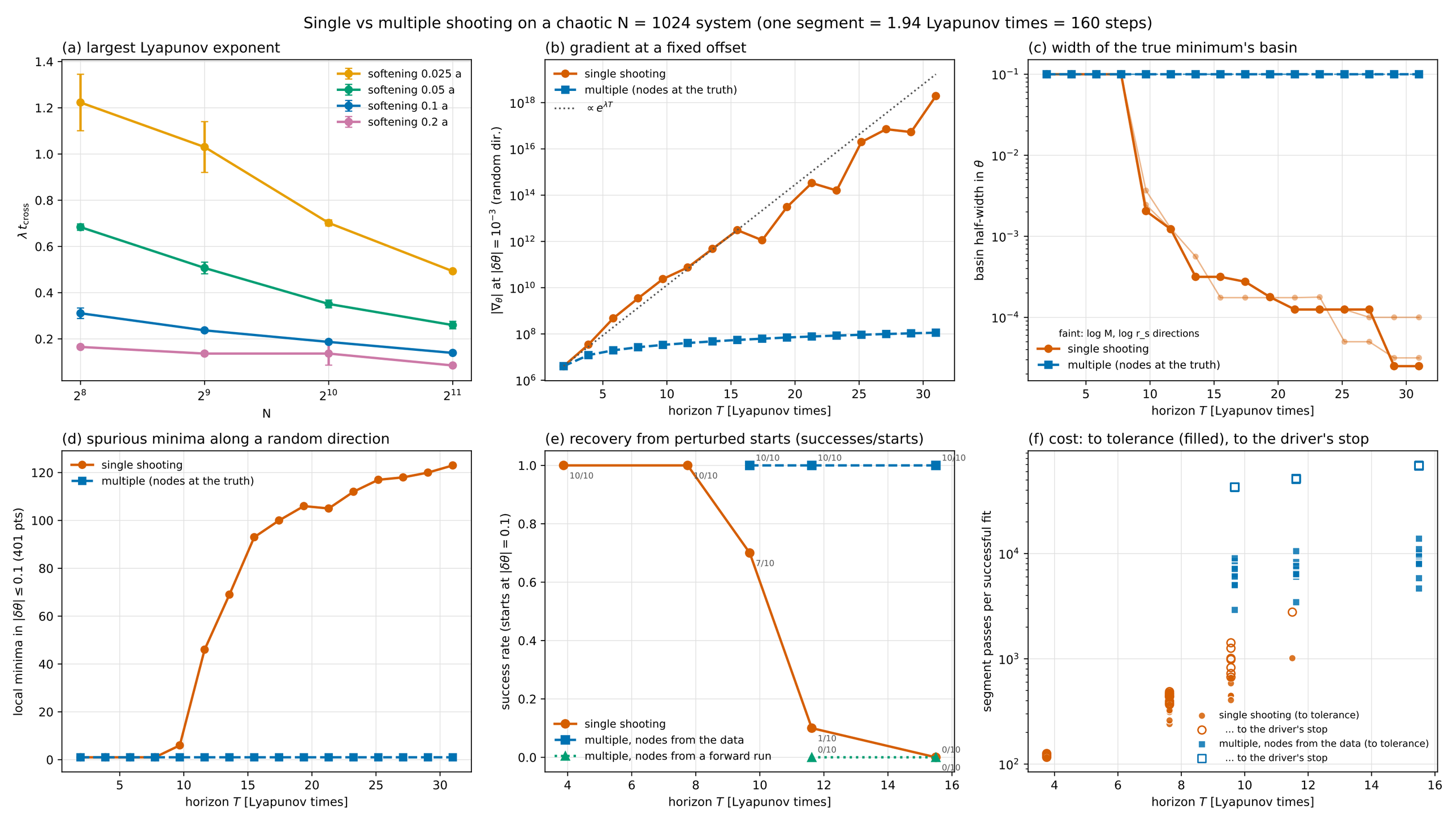
Single vs multiple shooting on a chaotic N = 1024 system (one segment = 1.94 Lyapunov times = 160 steps)
(a) largest Lyapunov exponent
softening 0.025 a
softening 0.05 a
softening 0.1 a
softening 0.2 a
$\lambda\, t_{cross}$
N
(b) gradient at a fixed offset
single shooting
multiple (nodes at the truth)
$\propto e^{\lambda T}$
$|\nabla_\theta|$ at $|\delta\theta| = 10^{-3}$ (random dir.)
horizon T [Lyapunov times]
(c) width of the true minimum's basin
basin half-width in θ
faint: log M, log r_s directions
single shooting
multiple (nodes at the truth)
horizon T [Lyapunov times]
(d) spurious minima along a random direction
single shooting
multiple (nodes at the truth)
local minima in $|\delta\theta| \le 0.1$ (401 pts)
horizon T [Lyapunov times]
(e) recovery from perturbed starts (successes/starts)
10/10
10/10
10/10
10/10
7/10
1/10
0/10
0/10
0/10
single shooting
multiple, nodes from the data
multiple, nodes from a forward run
success rate (starts at $|\delta\theta| = 0.1$)
horizon T [Lyapunov times]
(f) cost: to tolerance (filled), to the driver's stop
segment passes per successful fit
single shooting (to tolerance)
... to the driver's stop
multiple, nodes from the data (to tolerance)
... to the driver's stop
horizon T [Lyapunov times]